PES1UG24CS372
Riddhi Mundada

LAB 7: Fuzzing for Input Validation Bugs

Initial Run

```
riddhi@Riddhis-MacBook-Air Lab 7 % python3 processor.py
==== Running parse_season_pass_validity ====
Enter pass expiry (YYYY/MM/DD): 2026/12/12
Pass Expiry: 2026-12-12 (VALID)

==== Running compute_trail_gradient ====
Enter drop:distance (e.g. 250:1000): 250:2000
Gradient %: 12.5

==== Running calculate_lift_access_tier ====
Enter zone code (e.g. BEG, INT, ALL): INT
Lift Pass Price: $ 60.0
```

Failing Hypothesis Tests

```
(venv) riddhi@Riddhis-MacBook-Air Lab 7 % python3 -m pytest test_processor.py
============================= test session starts ==============================
platform darwin -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: /Users/riddhi/Desktop/sem5/SE/lab6/Lab 7
plugins: hypothesis-6.170.15
collected 3 items

test_processor.py FFF                                                    [100%]

=================================== FAILURES ===================================
_____ test_parse_season_pass_validity_no_crash _____

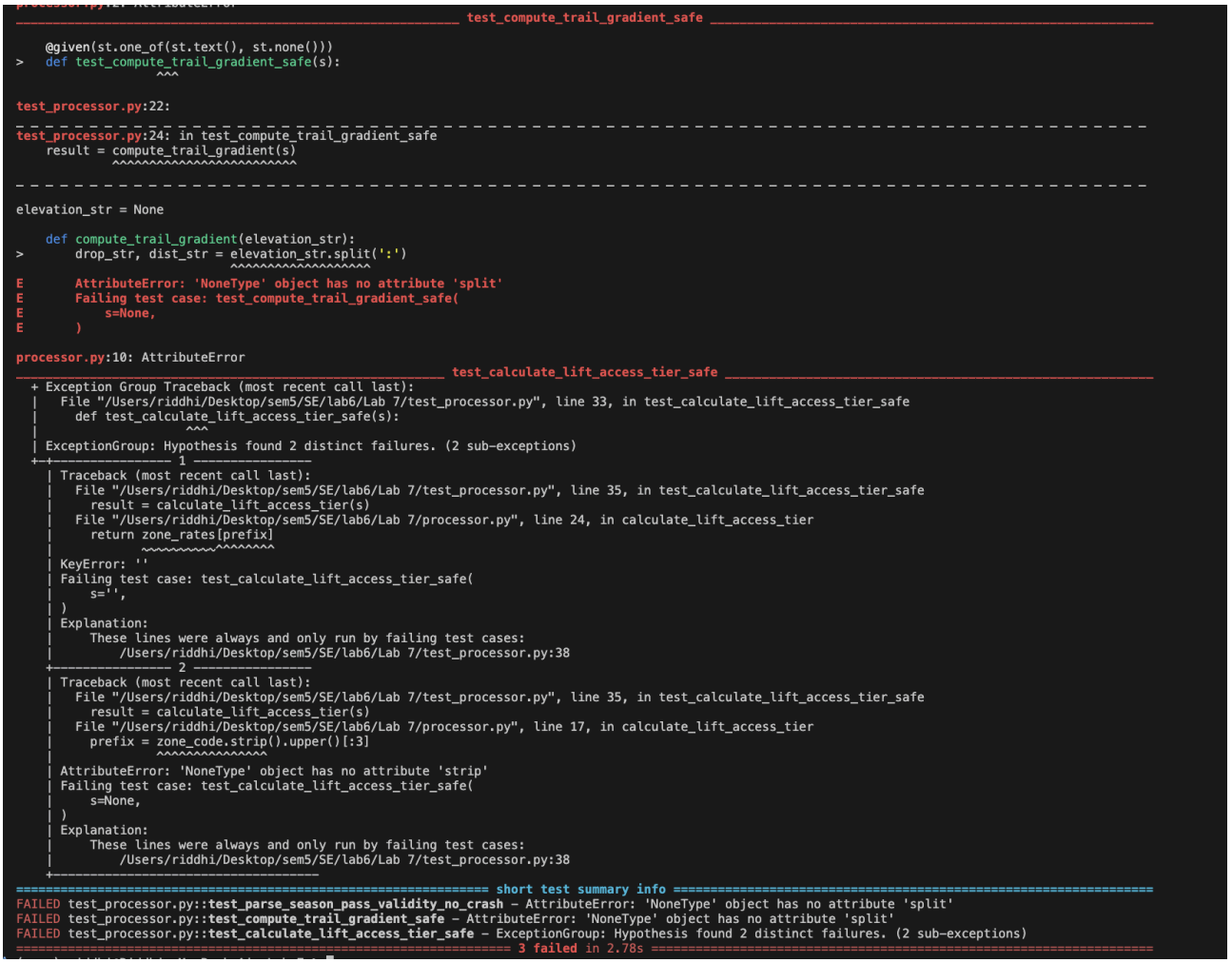
    @given(st.one_of(st.text(), st.none()))
>   def test_parse_season_pass_validity_no_crash(s):
                ^^^

test_processor.py:11:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
test_processor.py:13: in test_parse_season_pass_validity_no_crash
    result = parse_season_pass_validity(s)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

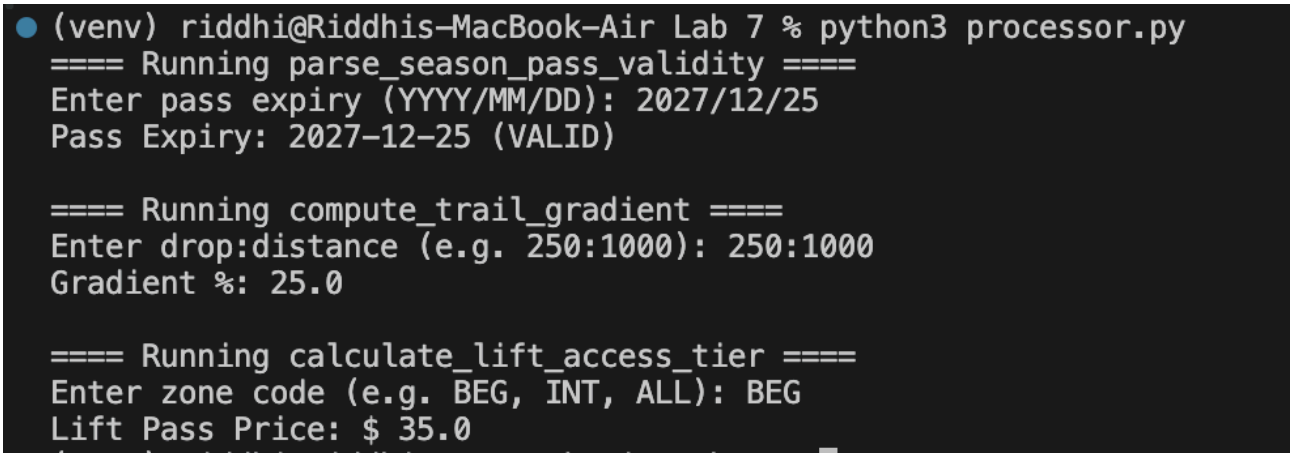
date_str = None

    def parse_season_pass_validity(date_str):
>       parts = date_str.split('/')
                ^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'
E       Failing test case: test_parse_season_pass_validity_no_crash(
E           s=None,
E       )

processor.py:2: AttributeError
```



Fixed Run



Successful Hypothesis Tests

